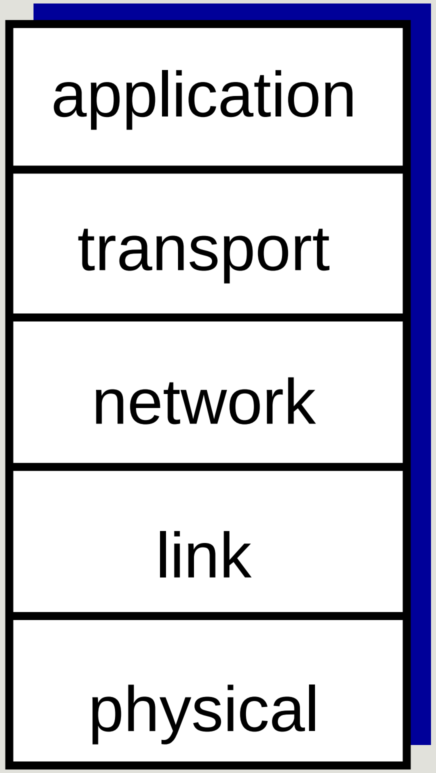
Application Protocols: DNS

FALL 2021

CSC311

PROF. EARL

The Internet Protocol (TCP/IP) Stack



A refresher on the five layers.

Application: Networked Applications
- HTTP, FTP, SMTP

Transport: Process-Process Data Transfer
- TCP & UDP

Network: Routing of Datagrams from source to destination
- IP and Routing Protocols

Link: Data Transfer between neighboring network elements
- Ethernet, 802.11 (WiFi), PPP

Physical: The bits “on the wire”

DNS

IP Address

- 127.0.0.1

“Name”

- localhost

Remembering names vs remembering numbers.

Example of mapping names and numbers?

DNS – Domain Name System

Distributed Database

- Many ***name servers*** hold various bits of information

Application Layer Protocol: Hosts, name servers communicate to resolve names (address <-> name)

- Worth Noting: Core Internet Function implemented as app-layer protocol
- Introduces complexity at Network Edge

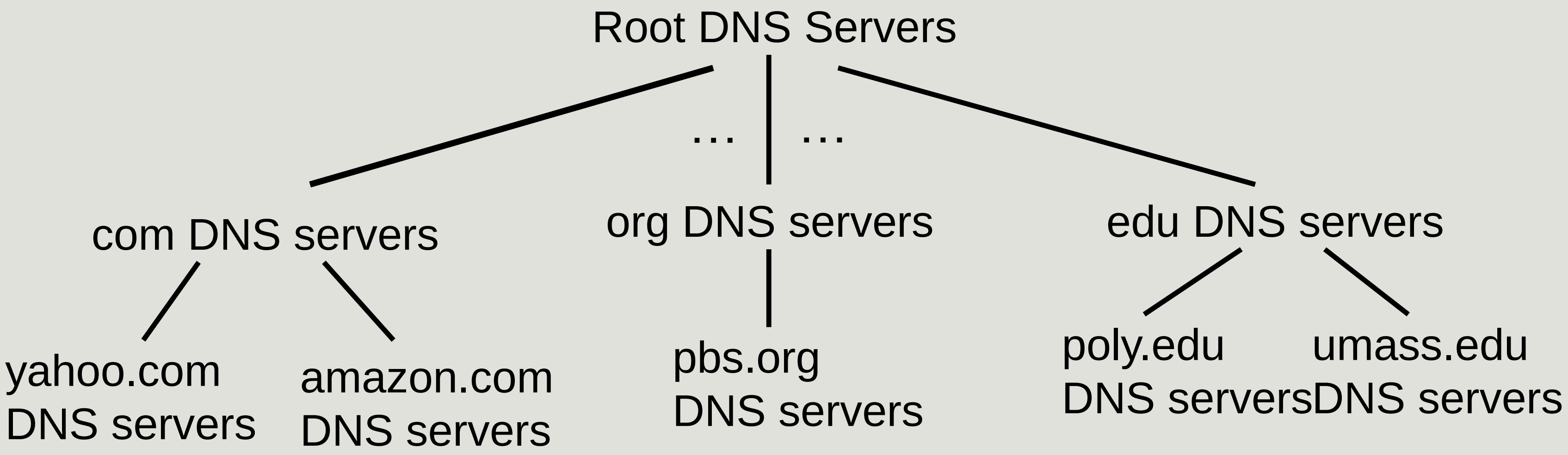
DNS: Services and Structure

Services:

- Hostname to IP address translation
- Host aliasing
  - canonical, alias names
- Mail server aliasing
- load distribution

Distributed, Hierarchical Database

- Root DNS
- tld DNS servers
- domain DNS servers



DNS Terminology

Domain Name – Human-friendly name used to associate internet address for a given resource

- i.e. google.com

Top-Level Domain (or TLD) – The general part of a domain. The part of the name found following the last dot "." at the right of the name. Common ones: com, edu, gov, org, io.

Hosts – The hosts within a given domain. Domain owners can define names for host's with in their organization. Typically webservers are found at **www.**example.com, FTP at **ftp.**example.com. There is no naming scheme here, as long as the names are unique.

Subdomains – An extension of a domain name. Typically used to group certain hosts, i.e for a CS department you would see www.cs.example.edu

- www still refers to the host, but www is a specific webhost for the CS department.

www.sci.pitt.edu

Read left-to-right: Most specific to least specific.

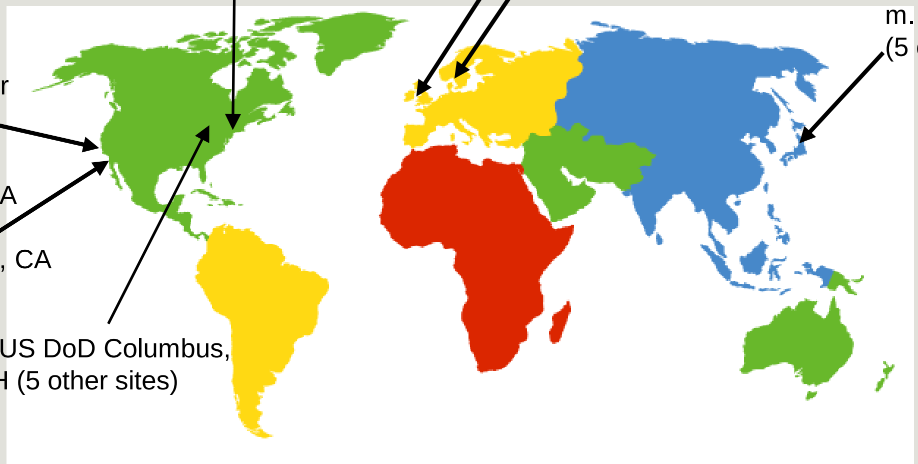
Source: DigitalOcean DNS Terminology Components and Concepts

Root Servers

Authoritative name servers – Network of hundreds of servers around the world

13 “Logical” servers split replicated on physical machines.

12 Independent Originations www.root-servers.org | https://www.iana.org/domains/root/servers



TLD, Authoritative Servers

Top-Level Domain (TLD) Servers:

- Responsible for com, org, edu, aero, jobs, etc.
- Individual Organization manages each TLD server
  - VeriSign Global Registry Services for .com
  - Educause for .edu
- https://www.iana.org/domains/root/db
- Knows IP address for organization’s Authoritative (or Domain-Level Name) servers.

Authoritative Servers

- Responsible for knowing IP – Name mappings hosts for an organization’s domain.
- Typically the final stop in a domain name lookup

Local DNS Servers

The Requester

- Typically found within an organization or ISP.
- Responsible for making domain requests on behalf of hosts located within their network

Default Name Server

Hosts making DNS query, send requests to their local DNS server.

- Default Name Server may have cache of recent requests
- If not, query gets forwarded into hierarchy of Root → TLD → Domain-Level Name Servers

DNS Name Resolutions

Iterative Queries:

- DNS Server gets queried and returns an answer without asking other DNS servers
- Usually points requester in another direction.

Recursive Queries:

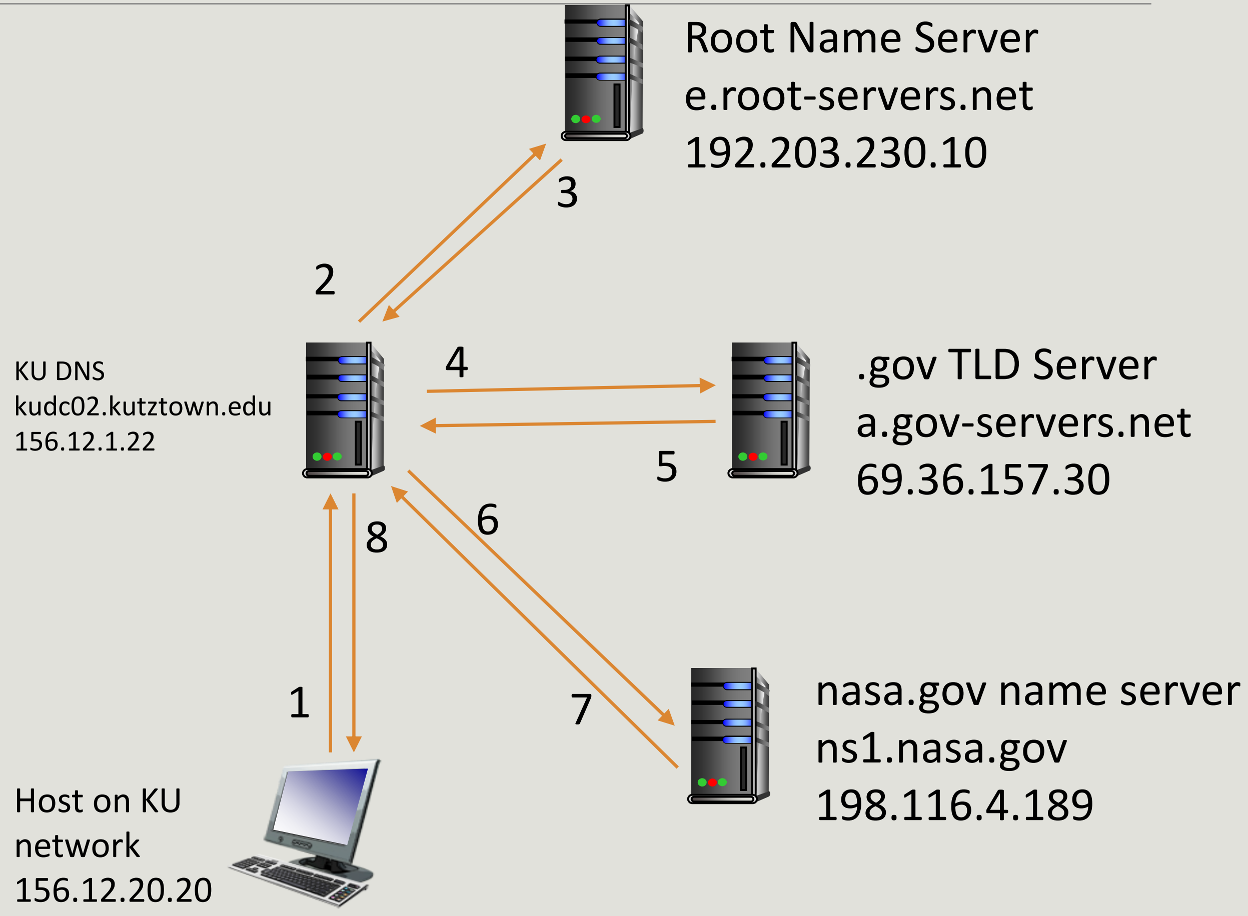
- DNS Server queries DNS servers until a definitive answer is returned to the requester. Queries made between servers are Iterative.
- Burden of resolution falls on contacted server.

DNS Resolution – Iterated

KU Host is attempting to get an IP address for www.nasa.gov

DNS Server doesn't have a cached copy of information, will need to reach out to root server.

Local DNS Server makes iterated requests to each server based on information received back.

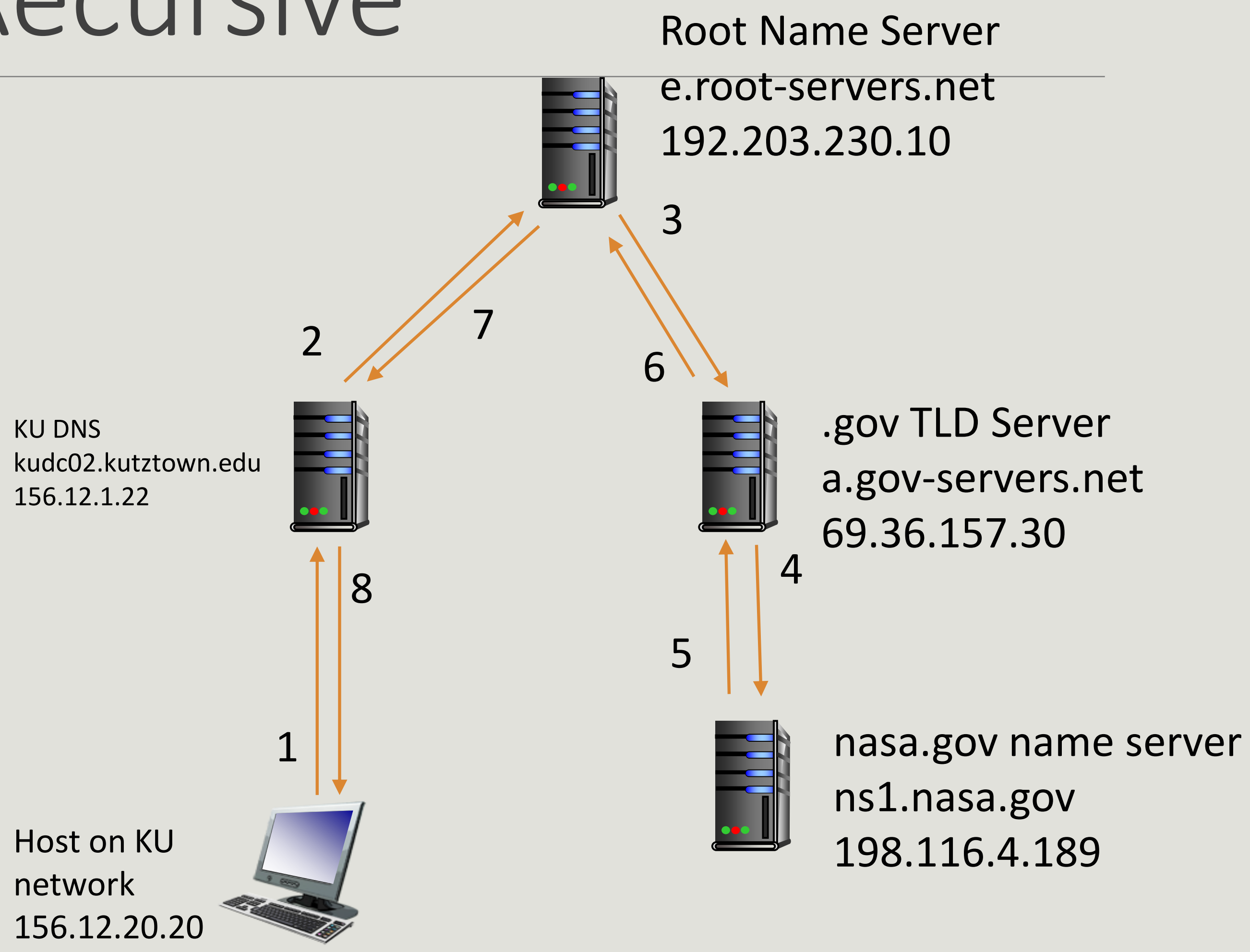


DNS Resolution – Recursive

KU Host is attempting to get an IP address for www.nasa.gov

DNS Server doesn't have a cached copy of information, will need to reach out to root server.

Each server will reach out to the next server to get the answer.



DNS Caching

Any nameserver that learns mapping, will cache it.

- TTL
- TLD Servers cached in local name servers
- Cached entries may go out of date

Quick Note: DNS Zones

Each step in the hierarchal structure is a “zone”

- “.” represents the Root Zone
- “.com” would be part of the TLD Zone
- “example.com” would be part of it’s owner’s zone (Including cs.example.com)

Each Zone is responsible for maintaining a name server.

- This server is the authoritative (trusted) record for that zone.

“.” zone is managed by those 13 organizations (Refer to the “Root Servers” slide)

“.com” is managed by VeriSign Global Registry Services

“example.com” is managed by the organization that owns the domain (or they delegate an external company to do so)

- cs.example.com could be part of the “example.com” zone or a separate zone

DNS “Whois” Lookups

DNS Records

DNS: Distributed database that stores *resource records* (RR)

RR format: `(name, value, type, ttl)`

Record Type	
A	Defines IP Address
CNAME	Canonical Name (An Alias)
HINFO	Returns Host Info (OS & CPU)
MX	Mail Exchange

Record Type	
AAAA	IPv6 Address
NS	Name Server (Specifies Authoritative name server for given domain

DNS Zone File

Most DNS Servers follow the BIND Format

- Berkley Internet Name Domain

Holds information about the domain's zone. Must start with a SOA record

- Start of Authority Record (More on that in the next slide)

Contains the RRs needed for the domain

name | ttl | record class | record type | record data

example.com.	IN	A	192.2.1.1
www	IN	CNAME	example.com.
mail	IN	A	192.2.1.2
mail2	IN	A	192.2.1.3

RFC 1035 Example - https://www.ietf.org/rfc/rfc1035.html#section-5.3

SOA Record

Structure:

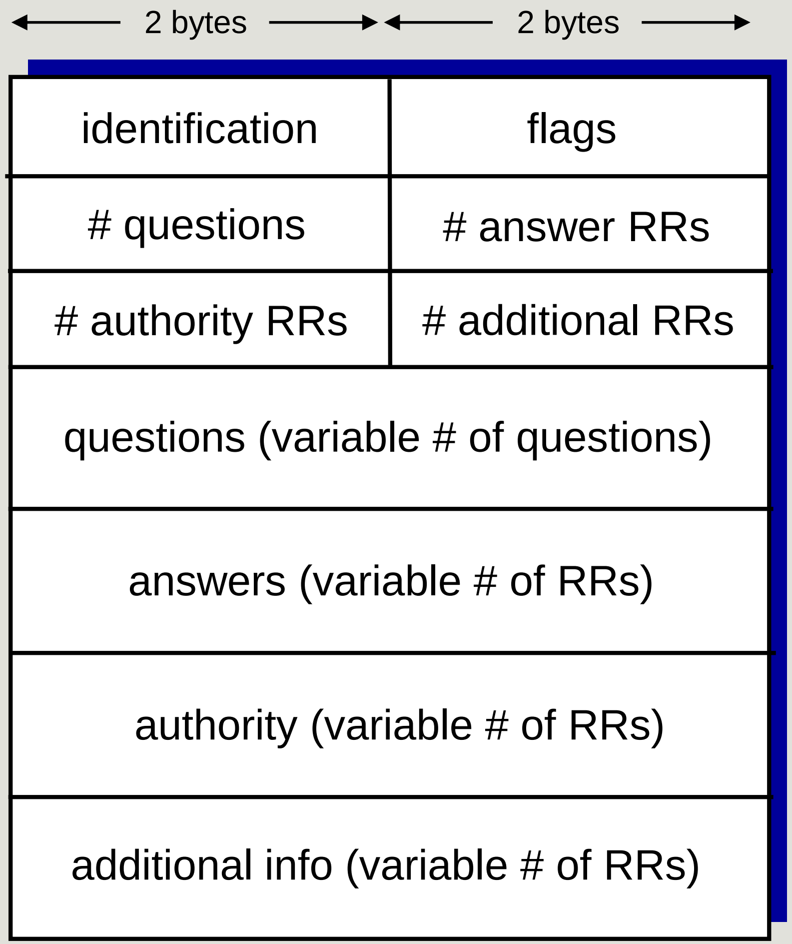
- name – The name of the zone (usually the second-level domain i.e. example.com)
- zone class – What type (Usually IN for internet)
- mname – Master Authoritative Name Server
  - i.e. ns.example.com
- Contact – The main contact for this domain
  - admin.example.com → Same as admin@example.com
- Record Data
  - Serial Number
  - Refresh
  - Retry
  - Expire
  - TTL
  - (All times in seconds)

DNS Protocol, Messages

Query and Reply messages with the same message format

Header:

- Identification: 16 bit num for query, replies to the query use the same num.
- flags
  - query or reply
  - recursion desired
  - recursion available
  - reply is authoritative



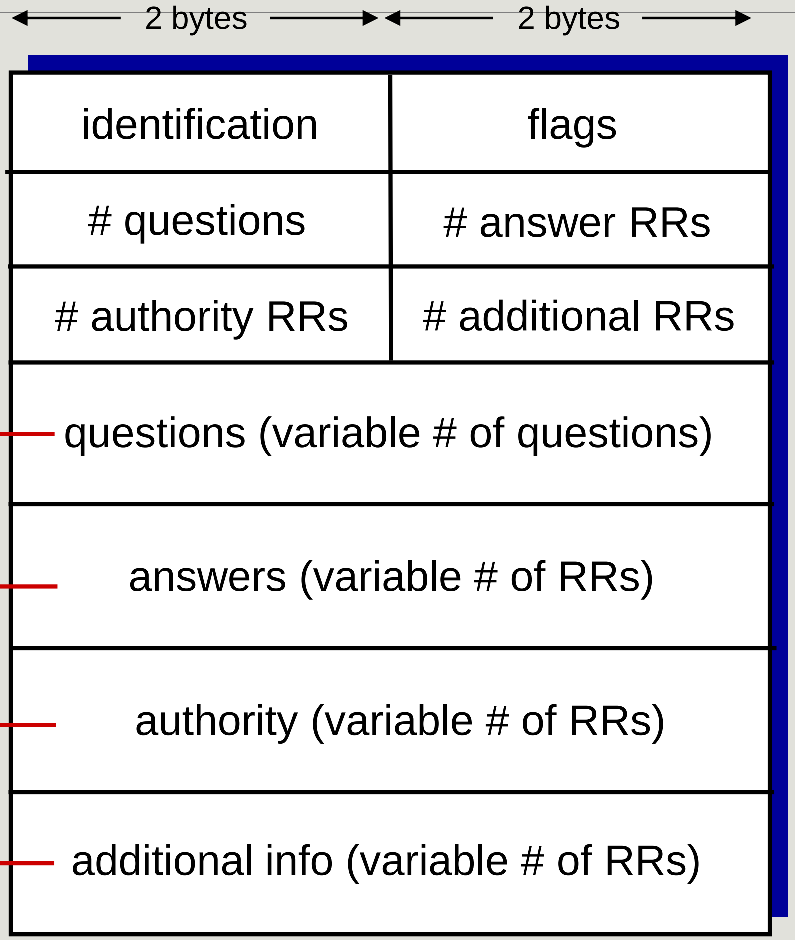
DNS Protocol, messages

name, type fields for a query

RRs in response to query

records for authoritative servers

additional "helpful" info that may be used



Notable Mentions: Pi-Hole

"A black hole for internet advertisements"

Can act as Default Name Server in your LAN

Maintains a list of domain names linked to advertisement companies

"Blocks" DNS queries for said domains

"Allows" DNS queries for others

Notable Mentions: Dynamic DNS

Enables “Dynamic” IP addresses to appear as a static DNS Mapping.

Residential ISPs typically change your public facing IP address

- Would require you to update the IP address every time.

Dynamic DNS uses a client to monitor IP address and does the updating itself.

Example: https://www.duckdns.org/